

CSE370 : Database Systems

Assignment 01/02/03 | Spring 2026

ID : <8 Digit Student ID> | Name : Full Official Name

No 1 Query (as Plain Text)	CREATE DATABASE Marathon_24301079; CREATE TABLE runners (student_id INT, student_name VARCHAR(50), email VARCHAR(60), distance_covered INT, time_taken_mins INT);
No 1 SS (of Query & Output in Shell)	<pre>MariaDB [(none)]> CREATE DATABASE Marathon_24301079; Query OK, 1 row affected (0.001 sec) MariaDB [(none)]> Use Marathon_24301079; Database changed MariaDB [Marathon_24301079]> CREATE TABLE runners (-> student_id INT, -> student_name VARCHAR(50), -> email VARCHAR(60), -> distance_covered INT, -> time_taken_mins INT ->); Query OK, 0 rows affected (0.010 sec)</pre>
No 2 Query (as Plain Text)	INSERT INTO runners VALUES (101, 'Mikasa Ackerman', 'mikasa.a@g.bracu.ac.bd', 5, 35),

	(102, 'Edward Elric', 'e.elric@iub.ac.bd', 10, 59), (103, 'Riza Hawkeye', 'riza.h@ewu.ac.bd', 5, 38), (104, 'Bruce Wayne', 'b.wayne@g.bracu.ac.bd', 10, 63), (105, 'Peter Parker', 'p.parker@iub.ac.bd', 5, 33), (106, 'Alphonse Elric', 'a.elric@iub.ac.bd', 10, 55), (107, 'Son Goku', 'son.goku@g.bracu.ac.bd', 5, 46), (108, 'Gon Freecss', 'gon.f@g.bracu.ac.bd', 10, 68), (109, 'Tanjiro Kamado', 'tanjiro.k@g.bracu.ac.bd', 5, 29), (110, 'Roy Mustang', 'roy.m@ewu.ac.bd', 10, 83);
No 2 SS (of Query & Output in Shell)	<pre> MariaDB [Marathon_24301079]> INSERT INTO runners VALUES -> (101, 'Mikasa Ackerman', 'mikasa.a@g.bracu.ac.bd', 5, 35), -> (102, 'Edward Elric', 'e.elric@iub.ac.bd', 10, 59), -> (103, 'Riza Hawkeye', 'riza.h@ewu.ac.bd', 5, 38), -> (104, 'Bruce Wayne', 'b.wayne@g.bracu.ac.bd', 10, 63), -> (105, 'Peter Parker', 'p.parker@iub.ac.bd', 5, 33), -> (106, 'Alphonse Elric', 'a.elric@iub.ac.bd', 10, 55), -> (107, 'Son Goku', 'son.goku@g.bracu.ac.bd', 5, 46), -> (108, 'Gon Freecss', 'gon.f@g.bracu.ac.bd', 10, 68), -> (109, 'Tanjiro Kamado', 'tanjiro.k@g.bracu.ac.bd', 5, 29), -> (110, 'Roy Mustang', 'roy.m@ewu.ac.bd', 10, 83); Query OK, 10 rows affected (0.031 sec) Records: 10 Duplicates: 0 Warnings: 0 </pre>
No 3 Query (as Plain Text)	SELECT * FROM runners WHERE distance_covered > 6;

No 3 SS
(of Query & Output
in Shell)

```
MariaDB [Marathon_24301079]> SELECT * FROM runners WHERE distance_covered > 6;
+-----+-----+-----+-----+-----+
| student_id | student_name | email | distance_covered | time_taken_mins |
+-----+-----+-----+-----+-----+
| 102 | Edward Elric | e.elric@iub.ac.bd | 10 | 59 |
| 104 | Bruce Wayne | b.wayne@g.bracu.ac.bd | 10 | 63 |
| 106 | Alphonse Elric | a.elric@iub.ac.bd | 10 | 55 |
| 108 | Gon Freecss | gon.f@g.bracu.ac.bd | 10 | 68 |
| 110 | Roy Mustang | roy.m@ewu.ac.bd | 10 | 83 |
+-----+-----+-----+-----+-----+
5 rows in set (0.000 sec)
```

No 4 Query
(as Plain Text)

```
SELECT UPPER(student_name) AS student_name FROM runners ORDER BY time_taken_mins DESC;
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No 4 SS
(of Query & Output
in Shell)

```
MariaDB [Marathon_24301079]> SELECT UPPER(student_name) AS student_name FROM runners ORDER BY time_taken_mins DESC;
+-----+
| student_name |
+-----+
| ROY MUSTANG |
| GON FREECSS |
| BRUCE WAYNE |
| EDWARD ELRIC |
| ALPHONSE ELRIC |
| SON GOKU |
| RIZA HAWKEYE |
| MIKASA ACKERMAN |
| PETER PARKER |
| TANJIRO KAMADO |
+-----+
10 rows in set (0.001 sec)

MariaDB [Marathon_24301079]>
```

No 5 Query (as Plain Text)	ALTER TABLE runners CHANGE COLUMN distance_covered total_distance DECIMAL(3,1);
No 5 SS (of Query & Output in Shell)	<pre> MariaDB [Marathon_24301079]> ALTER TABLE runners CHANGE COLUMN distance_covered total_distance DECIMAL (3,1); Query OK, 10 rows affected (0.034 sec) Records: 10 Duplicates: 0 Warnings: 0 MariaDB [Marathon_24301079]> SELECT * FROM runners; +-----+-----+-----+-----+-----+ student_id student_name email total_distance time_taken_mins +-----+-----+-----+-----+-----+ 101 Mikasa Ackerman mikasa.a@g.bracu.ac.bd 5.0 35 102 Edward Elric e.elric@iub.ac.bd 10.0 59 103 Riza Hawkeye riza.h@ewu.ac.bd 5.0 38 104 Bruce Wayne b.wayne@g.bracu.ac.bd 10.0 63 105 Peter Parker p.parker@iub.ac.bd 5.0 33 106 Alphonse Elric a.elric@iub.ac.bd 10.0 55 107 Son Goku son.goku@g.bracu.ac.bd 5.0 46 108 Gon Freecss gon.f@g.bracu.ac.bd 10.0 68 109 Tanjiro Kamado tanjiro.k@g.bracu.ac.bd 5.0 29 110 Roy Mustang roy.m@ewu.ac.bd 10.0 83 +-----+-----+-----+-----+-----+ 10 rows in set (0.001 sec) MariaDB [Marathon_24301079]> </pre>
No 6 Query (as Plain Text)	ALTER TABLE runners ADD COLUMN kmh DECIMAL(3,1);

No 6 SS (of Query & Output in Shell)	<pre>MariaDB [Marathon_24301079]> ALTER TABLE runners ADD COLUMN kmh DECIMAL(3,1); Query OK, 0 rows affected (0.005 sec) Records: 0 Duplicates: 0 Warnings: 0</pre>
No 7 Query (as Plain Text)	UPDATE runners SET kmh = (total_distance * 60) / time_taken_mins;
No 7 SS (of Query & Output in Shell)	<pre>MariaDB [Marathon_24301079]> UPDATE runners SET kmh = (total_distance * 60) / time_taken_mins; Query OK, 10 rows affected, 10 warnings (0.002 sec) Rows matched: 10 Changed: 10 Warnings: 10</pre>
No 8 Query (as Plain Text)	SELECT student_id, student_name, kmh, (total_distance / 10) * (time_taken_mins / 60) * 100 AS points FROM runners;

No 8 SS
(of Query & Output
in Shell)

```
MariaDB [Marathon_24301079]> SELECT student_id, student_name, kmh, (total_distance / 10) * (time_taken_
mins / 60) * 100 AS points FROM runners;
+-----+-----+-----+-----+
| student_id | student_name | kmh | points |
+-----+-----+-----+-----+
| 101 | Mikasa Ackerman | 8.6 | 29.166666650 |
| 102 | Edward Elric | 10.2 | 98.333333300 |
| 103 | Riza Hawkeye | 7.9 | 31.666666650 |
| 104 | Bruce Wayne | 9.5 | 105.000000000 |
| 105 | Peter Parker | 9.1 | 27.500000000 |
| 106 | Alphonse Elric | 10.9 | 91.666666600 |
| 107 | Son Goku | 6.5 | 38.333333300 |
| 108 | Gon Freecss | 8.8 | 113.333333300 |
| 109 | Tanjiro Kamado | 10.3 | 24.166666650 |
| 110 | Roy Mustang | 7.2 | 138.333333300 |
+-----+-----+-----+-----+
10 rows in set (0.000 sec)

MariaDB [Marathon_24301079]>
```

No 9 Query
(as Plain Text)

SELECT student_name FROM runners WHERE total_distance = 10 AND time_taken_mins BETWEEN 40 AND 60;

No 9 SS (of Query & Output in Shell)	<pre> MariaDB [Marathon_24301079]> SELECT student_name FROM runners WHERE total_distance = 10 AND time_taken_ mins BETWEEN 40 AND 60; +-----+ student_name +-----+ Edward Elric Alphonse Elric +-----+ 2 rows in set (0.000 sec) MariaDB [Marathon_24301079]> </pre>
No 10 Query (as Plain Text)	SELECT student_name FROM runners WHERE email LIKE '%@g.bracu.ac.bd';
No 10 SS (of Query & Output in Shell)	<pre> MariaDB [Marathon_24301079]> SELECT student_name FROM runners WHERE email LIKE '%@g.bracu.ac.bd'; +-----+ student_name +-----+ Mikasa Ackerman Bruce Wayne Son Goku Gon Freecss Tanjiro Kamado +-----+ 5 rows in set (0.000 sec) MariaDB [Marathon_24301079]> </pre>